

The Modelled Spread

A System of Equations for the COMEX–London Gold Basis,
and How to Read It

Methods note, GOLD project

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The spread between the COMEX gold futures price and the London benchmark is usually reported as a single annualised number. That number mixes three economically distinct objects: a premium for metal being in New York rather than London, a premium for the risk that a tariff lands before delivery, and the cost of carrying metal through time. This note writes the system down so the three can be held apart, fixes the units, runs one real date through every equation, states which combinations the data identify and which they do not, sets the restrictions the system imposes against their measured values, and lists the assumptions with the point at which each one fails.

Equation numbers match Section 9 of `SPREAD_EXPLAINED.html` and Section 4 of `spread_model_note.pdf`. Every figure quoted is printed by `spread_model_checks.py`, which reads only files already in `data/processed/`; its saved output is the audit trail for this document.

1 Notation and conventions

Symbol	Meaning	Status
S_t	London price, unallocated, 15:00 London	observed
$F_t(\tau)$	COMEX price for delivery in τ days	observed, 13:30 New York
$V_t(\tau)$	value today of metal deliverable in New York at τ	latent
p_t	location premium, fraction of spot (place, form, trust)	latent; estimated
r_t	dollar interest rate (SOFR; fed funds before Apr 2018)	observed
s^N	New York storage cost, per year	assumed; see §8
ℓ_t^N, ℓ_t^L	lease rate on metal in New York, in London	latent
h_t, θ_t	hazard rate of a tariff; tariff rate if it lands	latent
$\Lambda_t(\tau)$	probability a delivery at τ is tariffed	latent
κ_t	all-in cost of moving an ounce between locations	latent; estimated
ε_t	gold's return from 15:00 London to the settlement	latent; measurable
Q_t	metal shipped, tonnes a month	observed (Swiss customs)
$\bar{Q}$	refining and secure-freight capacity, tonnes a month	latent

Conventions the arithmetic depends on.

- *Subscripts and superscripts.* t indexes the trading date; superscripts N and L are locations, New York and London, never powers.
- *Rates* are annualised decimals: 0.05 is five per cent a year. Differences between rates are quoted in percentage points (pp).
- *Day count* is 365 throughout. SOFR is quoted on a 360-day year, so using it unadjusted

overstates carry by the factor 365/360 — about 0.06 pp at 2025 rates. Not corrected; small relative to everything here.

- τ is calendar days to first notice, the last business day of the month before the delivery month. First notice rather than last trade, because from that day a holder can be handed metal, so it is the date a financial holder must be out by.
- *Compounding.* Equations (1), (8) and the fitted curve are continuous; (2), (12) and (13) are simple. The difference is the compounding cross-term — 40 cents an ounce on the worked date in §6. Mixed conventions are a real if minor inconsistency in this system.
- $\hat{p}_t$ is a log difference, $\alpha_t - \ln S_t$, so it differs from the simple ratio P_t/S_t in the third decimal: 0.473% against 0.475% on the worked date. Either is fine; they should not be mixed within a table.
- *Signs.* $p_t > 0$ means New York is dear, so metal flows west; $p_t < 0$ means the reverse. A rise in the lease rate lowers the measured dislocation.
- *Prices* are US dollars per troy ounce; flows are tonnes a month (1 t = 32,150.7 oz).

2 The system

Pricing through time, within New York.

$$F_t(\tau) = V_t(\tau) \cdot \exp\left[\left(r_t + s^N - \ell_t^N\right) \frac{\tau}{365}\right] \quad (1)$$

A COMEX contract is metal in a New York vault at a future date, so its price is that metal's value today compounded at the cost of holding it: interest, plus storage, less the lease income the holder forgoes. This is the only equation with a time dimension, and it is the gold version of covered interest parity with ℓ^N in the role of the foreign interest rate.

Pricing across locations, and the policy risk inside it.

$$V_t(\tau) = S_t [1 + p_t + \Lambda_t(\tau) \theta_t] \quad (2)$$

$$\Lambda_t(\tau) = 1 - \exp\left(-h_t \frac{\tau}{365}\right) \approx h_t \frac{\tau}{365} - \frac{1}{2} \left(h_t \frac{\tau}{365}\right)^2 \quad (3)$$

New York metal is worth the London price plus a premium. The premium has a standing part p_t — location, bar form, and trust in a warrant versus a claim on a bank — and a policy part that depends on the delivery date, because a contract delivering in nine months is likelier to be caught by a tariff than one delivering next month. Equation (3) is where a “level” premium stops being a level, and it leaves two signatures: the policy term enters the *slope* of the curve, and because Λ is concave in the horizon it bends the curve down. The expansion needs $h_t \tau / 365 \ll 1$; if the hazard is large, Λ saturates and the policy term loads on the level instead, which turns out to matter in §7.

What arbitrage enforces: a band, not a point.

$$-\kappa_t^{N \rightarrow L} \leq p_t S_t \leq \kappa_t^{L \rightarrow N} \quad \text{unless capacity binds} \quad (4)$$

$$\kappa_t^{L \rightarrow N} = \underbrace{c_{\text{freight}} + c_{\text{ins}} + c_{\text{recast}}}_{\text{physical, per ounce}} + \underbrace{\left(r_t + \ell_t^L\right) \frac{d_{\text{transit}}}{365} S_t}_{\text{financing in transit}} \quad (5)$$

Arbitrage does not hold p_t at zero; it holds it inside a band whose edges are the cost of shipping each way. The band's width is therefore not a nuisance parameter — it is the transfer cost

the kink regression is trying to recover. Equation (5) is also the only place London's lease rate appears: shipping ties up metal and money for the days in transit, and a London squeeze raises that cost, widening the band rather than moving the COMEX curve.

Measurement: the two legs are not observed at the same instant.

$$\widehat{F}_t(\tau) = F_t(\tau) \cdot e^{\varepsilon_t}, \quad \mathbb{E}[\varepsilon_t] = \mu, \quad \text{Var}[\varepsilon_t] = \sigma^2 \quad (6)$$

The London benchmark is struck at 15:00 London and the COMEX settlement about three and a half hours later, so ε_t is gold's return over that window. Every contract settles at the same instant, so ε_t is *common* to all of them on date t : it shifts the whole curve without tilting it. That single property decides which results need a timing fix and which do not.

Flow, with a capacity ceiling.

$$Q_t = \min\{\bar{Q}, \beta \max(0, p_t S_t - \kappa_t)\} + u_t \quad (7)$$

Metal moves once the premium clears the cost of moving it, and no faster than refineries and secure freight allow. The ceiling is why a premium far above the threshold can persist for months without being an arbitrage failure.

3 What is estimated

Taking logs of (1) and (2), substituting (3) and (6), and collecting powers of τ gives a curve that is quadratic in the horizon. This is the regression run each day across that date's contracts, weighted by the square root of open interest:

$$\ln \widehat{F}_t(\tau_i) = \alpha_t + \beta_t \tau_i + \gamma_t \tau_i^2 + \eta_{it} \quad (8)$$

A contract enters if $15 \leq \tau_i \leq 400$ days, open interest is at least 1,000 lots, and a settlement exists: under 15 days a contract is in its delivery scramble, and far-dated or thinly held contracts carry stale settlements. That leaves a median of six contracts a day across 2,857 days, 2015–2026. The three fitted coefficients map onto the objects above:

$$\hat{p}_t \equiv \alpha_t - \ln S_t = p_t + \varepsilon_t \quad (9)$$

$$\hat{\omega}_t \equiv 365\beta_t - r_t = s^N - \ell_t^N + h_t \theta_t \quad (10)$$

$$\gamma_t = -\frac{h_t^2 \theta_t}{2 \cdot 365^2} \leq 0 \quad (11)$$

Read them as: the intercept is a *level*, in per cent of spot; the slope is a *rate*, in points a year; the curvature is a *bend*, and the only coefficient that responds to policy risk alone.

4 The modelled spread

$$b_t(\tau) \equiv F_t(\tau) - S_t = S_t \left[\underbrace{p_t}_{\text{place, form, trust}} + \underbrace{\Lambda_t(\tau) \theta_t}_{\text{policy risk}} + \underbrace{(r_t + s^N - \ell_t^N) \frac{\tau}{365}}_{\text{carry}} \right] \quad (12)$$

$$\text{disloc}_t(T) \approx p_t \frac{365}{T} + h_t \theta_t + (s^N - s) - \ell_t^N \quad (13)$$

Equation (12) is the spread itself. Equation (13) is what the project's headline number does to it: reading the fitted curve at a fixed horizon T and subtracting an assumed carry rate s

divides the location premium by the horizon, leaves the policy and lease terms alone, and adds whatever error sits in the storage assumption. The multiplier $365/T$ is 12.2 at thirty days, 4.06 at ninety and 2.03 at one hundred and eighty, so the horizon silently sets how much of the reported number is location and how much is carry. January 2025 reads 6.90 pp at thirty days, 2.83 at ninety and 1.84 at one hundred and eighty. That is the case for reporting $\hat{p}_t$ and $\hat{\omega}_t$ separately rather than any single dislocation.

5 Units and scales: how to read a coefficient

The three quantities live in different units, and most confusion in this literature is a failure to convert between them.

	% of spot	\$/oz at \$2,756	\$/oz at \$4,200
1 pp of disloc(T), $T = 30$	0.0822%	\$2.27	\$3.45
1 pp of disloc(T), $T = 90$	0.2466%	\$6.80	\$10.36
1 pp of disloc(T), $T = 180$	0.4932%	\$13.59	\$20.71
	% of spot	pp read at 90d	
\$1/oz of level premium	0.0363%	0.15	
\$2/oz, a typical transfer cost	0.0726%	0.29	
\$13.08/oz, the worked date	0.4745%	1.92	

Two reference magnitudes are worth carrying: the calm-period level is $-\$0.55$ an ounce, and the estimated flow threshold is $\$3.77$ an ounce with a 90% interval of $\$0.83$ – $\$10.25$. A one-point move in a 90-day dislocation is therefore worth roughly twice the entire transfer cost — which is why the level, and not the annualised rate, is the quantity a shipping decision responds to.

6 One day through the whole system: 29 January 2025

Equation	Quantity	Value
observed	S_t , London PM auction	\$2,756.30
observed	r_t , SOFR	4.3500%
assumed	s , storage	0.25%
(8)	$\exp(\alpha_t)$, the curve at $\tau = 0$	\$2,769.38
(8)	$365\beta_t$, the curve's slope	5.2165%
(8)	γ_t , the curve's bend	$-0.030 \times 10^{-6}/\text{day}^2$
(9)	$\hat{p}_t$, location premium	+0.473% of spot = +\$13.08/oz
(10)	$\hat{\omega}_t = s^N - \ell_t^N (+h_t\theta_t)$	+0.866 pp
(1),(2)	$F_t(90) = \exp(\alpha_t + 90\beta_t)$	\$2,805.23
(1),(2)	implied rate at 90 days	7.1991%
(12)	spread at 90 days	\$48.93/oz
	of which location	\$13.08
	of which carry	\$35.45
	of which compounding	\$0.40
(13)	disloc $_t(90)$	2.599 pp
	location, $\hat{p}_t \times 365/90$	1.919
	carry wedge, $\hat{\omega}_t - s$	+0.616
	cross-term	+0.063

Read down the table: of the \$48.93 spread that day, \$35.45 was the time value of money and storage, \$13.08 was New York being dear, and forty cents was compounding. The headline 2.60 pp dislocation is three-quarters location premium — but only because it was read at ninety days.

7 Identification: what the data pin down

Object	Identified by	Caveat
p_t	the intercept, $\alpha_t - \ln S_t$	carries ε_t ; a monthly mean has reliability 0.80
$s^N - \ell_t^N$	the slope, $365\beta_t - r_t$	not separable without an external storage or lease quote
$h_t\theta_t$	the slope in excess of a calm baseline	only as good as that baseline
h_t, θ_t apart	slope together with curvature	rejected in practice; see below
κ_t	the kink in (7), and the calm band of $p_t S_t$	wide interval, \$0.83–10.25
ε_t	intraday COMEX prices at the auction instant	not on disk; one Databento job
place / form / trust	<i>not identified</i>	needs warrant- or bar-level prices
ℓ_t^L	<i>not identified from this curve</i>	enters only through κ_t in (5)

The hazard and the tariff, in principle. Equations (10) and (11) are two moments in two unknowns, once a baseline for $s^N - \ell_t^N$ is fixed. Writing $A \equiv h_t\theta_t$ from the slope and $B \equiv -2 \cdot 365^2\gamma_t$ from the curvature,

$$h_t = \frac{B}{A}, \quad \theta_t = \frac{A^2}{B}.$$

In practice it is rejected, informatively. Taking the calm years 2015–19 as the baseline ($\hat{\omega} = 0.58\%$, $\gamma = -0.007 \times 10^{-6}$):

- For January 2025, $\hat{\omega} = +0.975$ pp gives $A = 0.00395$ a year. The curvature then implies $h_t \approx 3.4$ – 3.9 a year — a 57–62% chance of a tariff within ninety days — paired with $\theta_t \approx 0.10$ – 0.12% . A near-certain tariff of one tenth of one per cent is not a description of what was being discussed, so the over-identifying restriction fails. The curvature is mostly ordinary term structure: it is already negative in the calm years.
- Run the other way, the slope *bounds* the risk-neutral tariff expectation, and the bound is small. Imposing a 10% chance within ninety days gives $\theta_t = 0.92\%$; a 25% chance gives 0.34%; a 50% chance gives 0.14%. Against a tariff under discussion at 10–39%, the term structure was pricing almost none of it.
- Across the whole episode window, November 2024 to April 2025, $\hat{\omega}$ averaged $+0.561$ pp — *below* the calm baseline — so $A < 0$ and the slope prices no positive hazard at all.

The reading is not that the tariff scare was imaginary but that it was priced as a *level* on metal already in New York rather than as a term premium on future delivery. That is consistent with the level accounting for roughly 85% of the January 2025 move once the five-year calm mean rather than a single calm day is used as the counterfactual. Equation (3) earns its place as a restriction the data reject, not as a channel they support.

8 Restrictions, and what the data say

Estimated over 2,857 daily fits, 2015–2026:

Prediction	From	Measured	Verdict
The timing error loads on the intercept, not the slope	(6), (9), (10)	level on the day’s return +0.193 (s.e. 0.015); slope on it −0.012 (s.e. 0.011)	holds
Level noise is a 3.5-hour slice of gold’s variance	(6)	predicted 0.39% of spot; measured level s.d. 0.50%, so 61% of the variance	holds
Yesterday’s return says nothing about today’s level	(6)	+0.049 (s.e. 0.017)	small violation
In calm markets $ p_t S_t $ sits inside the transfer band	(4)	calm mean −\$0.55/oz against $\kappa \approx \$1.5\text{--}3.8$	holds
The curve bends down, $\gamma_t \leq 0$	(3), (11)	mean over s.e. −10.7 calm, −22.0 in the episode	holds, but present in calm too
The slope prices a positive expected tariff while the threat is live	(10)	Jan 2025 +0.40 pp; whole window −0.02 pp	fails at window level
(10) and (11) jointly give plausible h_t, θ_t	(3), (11)	$h \approx 3.5/\text{yr}$ with $\theta \approx 0.1\%$	rejected
Flow is flat below the threshold, then rises	(7)	threshold \$3.77 [\$0.83, \$10.25]; 2–3 t/month below, 64 t median above	holds

One further restriction is not a test but an arithmetic identity, and it fails. The calm-period slope gives $s^N - \ell_t^N = +0.58$ pp a year. With $\ell^N \approx 0$ in calm years that implies storage of about 0.58%, not the 0.25% assumed; if storage really is 0.25%, the implied lease rate is −0.33%, which nobody quotes. One of the two is mis-set, and because s enters (13) one-for-one, so is every reported dislocation *level*. Read levels in differences until this is resolved.

9 Assumptions, and where each one fails

- **Futures are treated as forwards.** Daily margining makes a futures price differ from a forward when interest rates covary with the gold price. The wedge is small at these horizons, but it lands in β_t — which is where the lease rate is read.
- **One representative New York price.** Registered and eligible warrants, and 100-oz against kilo bars, are all folded into p_t . The registered/eligible split is itself a signal the project tracks separately.
- **One lease rate per location, flat across τ .** If the lease curve is sloped, part of it appears as curvature and contaminates §7.
- **The curve is log-quadratic.** A third-order term would reallocate weight between β_t and γ_t , which is precisely the margin the hazard identification rests on.
- **$\sqrt{\text{open interest}}$ weights are ad hoc.** The defensible weight is inverse settlement-error variance, for which open interest is a proxy of unknown shape. In practice the heaviest contract carries a median 46% of the weight (middle half 43–51%), so the “constant maturity”

curve is less independent of the front month than the name suggests.

- **ε_t is assumed mean zero.** Only its variance has been measured. The drift lower bound is \$0.31 an ounce: negligible against a \$14 episode premium, not negligible against a $-\$0.55$ calm mean. One intraday pull settles it.
- **(7) is estimated on monthly data.** Shipments respond to the premium and also close it, so the coefficient is an equilibrium relation rather than the supply of arbitrage. It is descriptive; identification needs within-month timing or an event study.
- **Swiss customs is a partial flow measure.** It misses direct London–New York shipments of already-deliverable bars, and non-Swiss origins. If the missing flows respond to the same signal, β is attenuated and κ_t biased.

10 How to use it

1. **Report $\hat{p}_t$ in dollars an ounce and $\hat{\omega}_t$ in points a year.** No horizon, no annualising of a level, and both are directly comparable to quantities measured outside this project: transfer costs for one, lease rates for the other.
2. **Re-time the New York leg** with an `ohlcv-1m` pull priced at the London auction instant, then re-estimate. Until then, divide kink slopes by the 0.80 reliability.
3. **Pin down storage.** A quoted vault-and-insurance schedule breaks the s^N versus ℓ^N tie and fixes every level in the project at once.
4. **Test whether the premium is a level or a risk** by reading the fitted curve at 30, 90 and 180 days and comparing p_t/S_t across them: equal means a level, rising means a horizon-dependent risk.
5. **Use the dated resolutions as identification.** At resolution $h_t \rightarrow 0$, so the slope and curvature should step at the April 2025 exemption and the July 2025 customs ruling while the level does not.
6. **Re-specify (7) with the ceiling**, and say plainly that the coefficient is descriptive.

Sources: `claude/spread-explainer/spread_model_checks.py` and `worked_examples.py`, both reading `data/processed/`; the kink threshold is from `claude/mechanism-figures/estimate_kink.py`. Where this note disagrees with earlier project text, or with an earlier version of itself, it says so.